DR5: Proposition-Ranked Nominators Cannot Distinguish a Commitment from Its Realisations

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Abstract

A nominator that scores individual propositions cannot, by itself, decide whether a scientific community should delete a commitment D when D has more than one non-equivalent surface realisation in the corpus. The scores it produces are per-proposition; the object of interest — D — is a class of propositions. Any move from a set of per-proposition scores to a single "score for D" must import an external grouping function g that says which propositions realise D. Once g is imported, the ranking is no longer proposition-based but class-based. We state and prove this as the **DR5 theorem**.

The theorem has a specific empirical shape and a general one. The specific shape: the DCR1c–DCR1f arc encountered the situation directly on a real pre-1905 electrodynamics corpus, where the target commitment T1 (absolute simultaneity) is present in *at least six* non-equivalent surface registers — Newton's metaphysical assertion, Larmor's mathematical rewrite $t \to t - vx/c^2$, Lodge's "definite and motion-independent duration," Maxwell's field-theoretic "instant across the whole medium," Poincaré's "regarded as simultaneous" convention, and a literary "duration is intrinsic" register surfaced by a held-out sanity set. Every matcher tuned to one register missed the others.

The general shape is a limit theorem about verification itself. In any setting where an oracle is used to check whether an agent has identified some latent commitment — did it find the requirement, or did the prompt install it? — the projection-vs-genuine-signal ambiguity is irreducible whenever the commitment has multiple non-equivalent surface forms. The DCR arc happens to make that irreducibility visible on a corpus with ground truth attached (Einstein's actual 1905 deletion), under a preregistered protocol that ruled out post-hoc matcher tuning, so the outcome is not a matcher-designer's error. It is what the theorem says will happen.

Companion empirical run (DCR2a) confirms the specific shape: even class-based scoring, applied to the DCR1e presupposition-extracted consensus, ranks T1 at position 3 at the 1904 target cut but *also* at position 2 at the 1880 deep placebo cut, because a single misfiring realisation is enough to inflate a low-cardinality class. Class-aware scoring is a strict generalisation of proposition-ranking, but it does not, by itself, eliminate the ambiguity. What it does is shift the locus: the ambiguity is now about whether the grouping function g is sound and complete for D, not about whether the ranking N found the right proposition.

1. Setup and definitions

Let C be a **corpus**: a finite set of propositions $P = \{p_1, \dots, p_n\}$. Propositions are the atomic units of assertion available to the community; the corpus is closed under whatever textual pre-processing extracts them.

A **commitment** D is a semantic property of propositions. A proposition p *realises* D iff p asserts or presupposes D. Whether p realises D is a fact about the content of p and the content of D, not about p's syntax, provenance, or membership in any subset.

Define the **realisation set** $$ \mathrm{realisations}(D, C) = \{p \in C : p \text{ realises } D\}. $$ This set may have any cardinality. We are interested in the case $|\mathrm{realisations}(D, C)| = k > 1$, where the commitment appears in more than one non-equivalent surface form. Call each such p a **realisation of D**, written r_i.

A **proposition-ranking nominator** is a function $N: P \to \mathbb{R}$ that assigns a score $N(p)$ to each proposition. We take the defining property of proposition-ranking to be that

$N(p)$ depends only on p: formally, N is invariant under permutations of $P \setminus \{p\}$. Equivalently, N has no access to which other propositions are in C, and no access to any grouping of P. This class includes every matcher studied in DCR1c–DCR1f — regex, keyword patterns, learned classifiers scoring each proposition independently — and it also includes the per-proposition scoring functions used across DR1–DR4.

N is **deletion-competent** for D iff there exists a ranking rule (top-1, top-k, or a fixed threshold) applied to the multiset $\{N(p) : p \in C\}$ such that the resulting output "identifies D" in some operational sense — e.g., the top-1 proposition realises D; the top-k contains a realisation of D; the propositions above threshold are exactly the realisations of D.

A **grouping function** is a map $g: P \to 2^P$ (or equivalently a partition of P into classes) such that g can identify $\mathrm{realisations}(D, C)$ as one of its classes. g is external structure: it is not derivable from the values of N alone.

2. Statement of the theorem

Theorem (DR5). *Let D be a commitment with $|\mathrm{realisations}(D, C)| = k > 1$, and let $N: P \to \mathbb{R}$ be a proposition-ranking nominator. Then:*

(a) In general, N assigns k distinct scores $\{N(r_1), \dots, N(r_k)\}$ to the realisations of D.

(b) No aggregation of $\{N(r_i)\}_{i=1}^{k}$ into a single "class score for D" is derivable from N alone; any such aggregation requires an external grouping function g that identifies $\mathrm{realisations}(D, C)$.

(c) Consequently, N cannot answer the question "is D ranked highly?" without importing g. Once g is imported, the ranking is not proposition-based but class-based.

3. Proof

(a) The realisations $r_1, \dots, r_k$ are, by hypothesis, non-equivalent surface forms — distinct propositions differing in wording, formalism, or presuppositional structure. N is a function on propositions, so in general $N(r_i) \neq N(r_j)$ for $i \neq j$. Coincidental equality is possible on measure-zero cases, but nothing in N's definition forces it.

(b) Suppose, for contradiction, that there is a function A such that $A(N, D, C)$ returns a single score for D, using only the values N takes on C. To evaluate A, one must at some point select the subset $\{N(r_1), \dots, N(r_k)\} \subseteq \{N(p) : p \in C\}$ on which to aggregate. Selecting this subset is exactly the operation of identifying $\mathrm{realisations}(D, C)$ within P. This selection is not a function of N's output, because N's output is a bag of real numbers indexed by propositions with no semantic tag for D. It requires a further map g that, given p, decides whether p realises D. This g is the grouping function.

(c) By (a), no single score is available from N. By (b), constructing one requires g. Therefore any procedure that answers "is D ranked highly?" reads g's output — the class of propositions realising D — and only then reads N. The primitive over which ranking is defined is now the class, not the proposition. $\square$

4. The DCR corollary and the empirical companion

Corollary. In the DCR arc, T1 (absolute simultaneity) has $k \geq 6$ non-equivalent surface realisations in the pre-1905 corpus. No proposition-ranking matcher can be simultaneously deletion-competent for T1 across all six registers; every choice of matcher pattern picks a subset of the realisation set and ranks the others as if unrelated. DCR1c–DCR1f's "*the matcher fires on one register at the expense of another*" is exactly the theorem operating.

Empirical companion (DCR2a, 2026-07-27, same day as this paper). Class-based scoring was implemented over the DCR1e presupposition- extracted consensus using the target_v4 matcher as the grouping function g. All three preregistered aggregation rules (cardinality, coverage, spread) rank the T1 class at position 3 at the 1904 target cut — a **one-position improvement** over the best T1 realisation's rank in a proposition-blind baseline (position 4). N3 GO.

But the same rules rank T1 at position **2** at the 1880 deep placebo cut, because Maxwell's instant across whole medium is the single T1 realisation there and low-cardinality classes with any hit at all dominate the cardinality-based aggregation. N4 NO_GO. DCR2a's overall verdict is NO_GO, with the licensed reading

> *class_scoring_fails_placebo: T1 outranks other classes at 1880 under*
>
> *some rule, meaning cardinality-1 hits inflate the class rank.*
>
> *Aggregation rule must require multi-document coverage.*

This is *not* a refutation of DR5. It is a specification of what DR5 requires. Class-aware nomination adds g (the matcher-derived class assignment); the theorem says that eliminates the multi-realisation ambiguity for well-behaved classes. But when g produces low-cardinality classes with borderline members, a second problem appears: **the correctness of g itself becomes the load-bearing question**, and the placebo-vs-projection ambiguity is inherited from N's wall to g's soundness/completeness. The class-scoring substitution is a real generalisation, but it moves the ambiguity — it does not delete it.

5. The general shape: a limit theorem about verification

The DCR arc's specific finding — that Einstein's deletion is a class, not a proposition, and no single-realisation matcher can catch it — generalises to a claim about verification of agent outputs at large.

Consider the abstract setting: an agent $\mathcal{A}$ is asked to identify a latent commitment D from a document (or reasoning trace). An oracle $\mathcal{O}$ checks whether $\mathcal{A}$'s output corresponds to D. In practice, $\mathcal{O}$ is often a matcher, a classifier, or an LLM judge — some function that scores individual propositions (sentences, program tokens, reasoning steps) and reports whether the target has been identified.

Two failure modes arise:

$\mathcal{A}$ found D but because the oracle was primed to find D in any output of the right surface shape.

should not be present, revealing the oracle is pattern-matching something other than D.

- **Projection.** The oracle fires on $\mathcal{A}$'s output not because
- **Placebo firing.** The oracle fires on a control input where D

The standard diagnostic — check the oracle on a placebo; if it fires, the oracle is projecting; if it stays silent, trust the positive result — assumes D has one surface form the oracle can recognise. **When D is a multi-realisation commitment, this diagnostic can be irreducibly ambiguous:** the placebo firing could indicate projection *or* a realisation of D the oracle-designer did not anticipate at the placebo cut.

DCR1f's Maxwell 1865 hit is exactly this. Under the DCR1f preregistered decision table, the 1880 hit at the deep placebo cut licenses **Reading B (extractor projecting)**. But the specific hit is a genuine presupposition of Maxwell's field theory — an "instant" over a spatially extended field — and by DR5's Corollary is a legitimate realisation of T1. The projection reading and the "placebo was never valid because a field theory already presupposes T1" reading are formally indistinguishable at the oracle level. Distinguishing them requires knowledge $\mathcal{O}$ does not have access to, and cannot acquire without importing a g that the DR5 theorem says was already necessary.

Practical consequence. For any agent-output-verification protocol where the target admits multiple surface forms — LLM reasoning verification, code-generation correctness, retrieval-augmented factual grounding, presupposition-detection, latent-goal identification — placebo-based projection checks can fail to distinguish projection from signal. The failure is not a design defect. It is a corollary of DR5. Any such protocol either requires an $a\text{-}priori$ grouping function g (which shifts the burden of correctness from the ranker to g) or requires machinery beyond proposition-independent scoring.

The DCR arc is, to our knowledge, the cleanest concrete instance of this pattern with ground truth attached — Einstein's actual 1905 deletion is known; the DCR1c preregistration ruled out post-hoc matcher tuning; the DCR1f held-out validation was generated by blind subagents at a fixed digest before target_v4 was drafted. The pattern generalises; the ground truth is the leverage that lets us see it.

6. What DR5 does not claim

DR5 does not claim that class-aware nomination is easy or well-defined; only that it is *necessary* for commitments with $k > 1$.

It does not claim g must be a matcher. g could be any function identifying realisations — a presupposition-inferring extractor (as DCR1e attempted), a learned classifier, a hand-curated ontology, or a canonical-form transformation applied at extraction time. Each has its own soundness/completeness questions.

It does not resolve DCR1e/f's B1 versus B2 (extractor projecting vs placebo invalid); those are separate empirical questions about whether a given g is sound at the placebo cut.

It does not require k to be known exactly; the theorem holds whenever $k > 1$, and the empirical companion DCR2a offered a lower bound on k for T1 in the pre-1905 corpus without needing an exact count.

It does not prescribe *which* class-aware nomination scheme to use. Cardinality, coverage, spread, and various IR-style aggregations are all compatible with the theorem's structural claim. The DCR2a run tested three and found all three insufficient at the placebo. Others may work; the theorem does not settle the question empirically.

7. Open questions

and complete for D? Soundness (every $p \in g(D)$ realises D) and completeness (every realisation of D in C is in $g(D)$) are the obvious candidates, but each is separately non-trivial. Constructing g empirically (as DCR1e/f attempted) is precisely the exercise DCR1f's precision-recall wall showed to be over-constrained.

extraction step *implicitly* define g? DCR1e's presupposition- inferring extractor arguably rewrites distinct surface forms into a canonical presuppositional statement before matching. If so, part of g's work is being done at extraction time, before the matcher sees the data. The DCR1f placebo firing at Maxwell suggests this is not yet fully deliberate.

Spencer's candidate-selection circularity one level up: if g is trained on labels derived from the same source as the ranking signal, the wall re-appears. This is the same argument that killed COGR Wave 1a; DR5 says it operates at the class-identification level too.

propositions but generalises to any C of items and any D admitting multi-form realisations. Code correctness, protocol compliance, and constraint satisfaction each admit multi-form realisations of "the correct implementation," and each should be susceptible to the same wall.

- **Correctness conditions on g.** Under what conditions is g sound
- **Extractor-implicit grouping.** Under what conditions does an
- **Learning g from data.** Can g be learned? A learned g risks
- **Non-linguistic verification.** DR5 was stated over corpora of

8. Relation to prior program findings

DR2 precedent. DR2 proved a two-nominator claim unreachable under its original cost definition. Progress required redefining cost so the primitive was no longer the extension whose cost was being counted. DR5 is parallel: it proves a class of nomination questions is unreachable under the proposition-ranking primitive, and progress requires redefining the primitive — here, from proposition to class. Both are structural walls, converted by proof, that admit the same rescue: the primitive was wrong, and once it is restated the corresponding empirical question becomes tractable.

Spencer's candidate-selection circularity, generalised. COGR Wave 1a died of a candidate-selection leak: the ranker was fed candidates constructed with knowledge of what would rank highly. DR5 says the leak has a formal analogue at the class-identification level: any grouping function g trained on the ranker's own signal collapses back into a proposition-ranking problem. Class-aware nomination is only a generalisation to the extent that g is *independent* of N.

The DCR1c–DCR1f arc. DCR1c through DCR1f can be re-read as three questions:

removal, relational residue, and provenance-cleared source vehicles.)

register.)

commitment? (DCR1e/f: no, on this corpus, with these instruments; and DR5 shows the "no" is structural for any such combination when $k > 1$.)

1. Can the extractor be honest? (DCR1: yes, with structural chrome
2. Can the matcher be validated? (DCR1d: yes, on Newton's explicit
3. Can extractor + matcher together identify a multi-realisation

The DCR arc has not proved deletion-repair nomination is unworkable in general. It has shown that its formulation as proposition-ranking hits a structural wall on the specific class of commitments Einstein deleted was drawn from. The wall is not a bug; it is what a well-run empirical program looks like when it correctly hits the limit of its primitive.

Appendix: reproduction

DR5 is a theorem paper; there is no code artifact. The empirical companion DCR2a reproduces via:

```
uv run --no-sync python -m experiments.date_cut_retrodiction.run_dcr2a
```

Its verdict at experiments/date_cut_retrodiction/results/dcr2a_verdict.json supplies the numbers cited in §4.

The DCR1c–DCR1f arc that motivates DR5 reproduces via each paper's own run_dcr1*.py. All prior verdicts are byte-identical to their published numbers; DR5 does not edit any prior module.

Preregistration. DR5 has no preregistration: it is a theorem paper, not an empirical run. The empirical companion DCR2a has one, at DCR2A_PREREGISTRATION.md, SHA-256 76d95494ca8968814a062f0b6bd9ceb287fcb3c135fe8647186b387a87a96717.